

Assignment 7:

Objective:

- Configure **dynamic routing using RIP Version 2**
- Configure **link-state routing using OSPF**
- Understand routing behavior and path selection
- Verify communication between networks

ROUTER 1:

```
enable
configure terminal
interface gig0/0/0
ip address 100.1.1.2 255.255.255.0
clock rate 64000
no shutdown
exit

interface gig0/0/1
ip address 20.1.1.1 255.255.255.0
no shutdown
exit

router rip
version 2
no auto-summary
network 100.1.1.0
network 20.1.1.0
exit

router ospf 1
router-id 1.1.1.1
network 100.1.1.0 0.0.0.255 area 0
network 20.1.1.0 0.0.0.255 area 0
end
```

ROUTER 2:

```
enable
configure terminal
interface gig0/0/0
ip address 100.1.1.1 255.255.255.0
no shutdown
exit

interface gig0/0/1
ip address 172.16.2.1 255.255.0.0
no shutdown
exit

router rip
version 2
no auto-summary
network 100.1.1.0
network 172.16.0.0
exit

router ospf 1
router-id 2.2.2.2
network 100.1.1.0 0.0.0.255 area 0
network 172.16.0.0 0.0.255.255 area 0
end
```

ROUTER 3:

```
enable
configure terminal
interface gig0/0/0
ip address 20.1.1.2 255.255.255.0
no shutdown
exit
```

```
interface gig0/0/1
ip address 10.2.2.1 255.255.255.0
no shutdown
exit
```

```
router rip
version 2
no auto-summary
network 20.1.1.0
network 10.2.2.0
exit
```

```
router ospf 1
router-id 3.3.3.3
network 20.1.1.0 0.0.0.255 area 0
network 10.2.2.0 0.0.0.255 area 0
end
```

Network Topology:

